Optimizing Public EV Charging Infrastructure in Baku

A geospatial coverage and network-based site-selection analysis

Technical report · analysis snapshot 2026-07-12 · OpenStreetMap road and charger data, official district polygons, and WorldPop 2026

Executive summary

This study evaluates how quickly locations across Baku's 12 land-clipped administrative districts can reach an inclusive set of public or access-unknown OpenStreetMap EV charging stations by car. Directed shortest paths are calculated on a static OSM driving graph. A population-weighted maximum-coverage model then selects 10 screening-feasible host sites from mapped parking, fuel, retail, transport, and public-facility points of interest.

Indicator	Baseline	After 10 sites	Change
Land within 15 minutes	33.3%	67.8%	+34.5 pp
Population within 15 minutes	84.1%	99.2%	+15.1 pp
Population within 10 minutes	58.8%	76.7%	+17.9 pp
Population-weighted p90 time (capped 30 min)	17.7 min	11.8 min	-5.8 min

The lowest baseline population accessibility is observed in Pirallahı (0.0% within 15 minutes). The recommended coordinates are planning shortlists, not construction approvals; grid capacity, land control, traffic access, safety, permits, and charger demand must be validated in the field.

Interpretation note: land percentages answer 'how much territory'; WorldPop percentages answer 'how much modelled resident population.' Neither measures charger uptime, queues, connector compatibility, charging power, or live congestion.

1. Study area and data

The study frame is the union of all 12 Baku district polygons published by Azerbaijan's official IDDA Open Data Portal, intersected with the Natural Earth 1:10m land layer so Caspian Sea territory is not counted in the denominator. The fixed OSM relation IDs remain in the manifest as an independent crosswalk.

Input	Source and role	Snapshot / license note
District boundaries	Azerbaijan IDDA official Regions of Azerbaijan GeoJSON; 12 features with PARENT_ID=10	Portal updated 2025; catalog declares Creative Commons Attribution
Existing chargers	OSM amenity=charging_station and man_made=charge_point via Overpass; private/no excluded	ODbL 1.0; inclusive unknown-access scenario
Driving network	OSM motorcar-accessible directed graph via OSMnx; edge speed from maxspeed/road-class defaults	Static model, not live traffic
Population	WorldPop 2026 constrained 100 m people-per-pixel raster; aggregated to analysis cells	R2025A alpha; DOI 10.5258/SOTON/WP00839
Land mask	Natural Earth 1:10m land polygons	Public domain

Station inclusion and data quality

The inclusive baseline retains 18 OSM station sites after spatial clipping, explicit private/no filtering, and deduplication of standalone charge-point objects near mapped charging-station sites. Missing access tags are treated as potentially public and identified as unknown, not asserted to be public. OSM records are incomplete and should be cross-checked against operators before policy use.

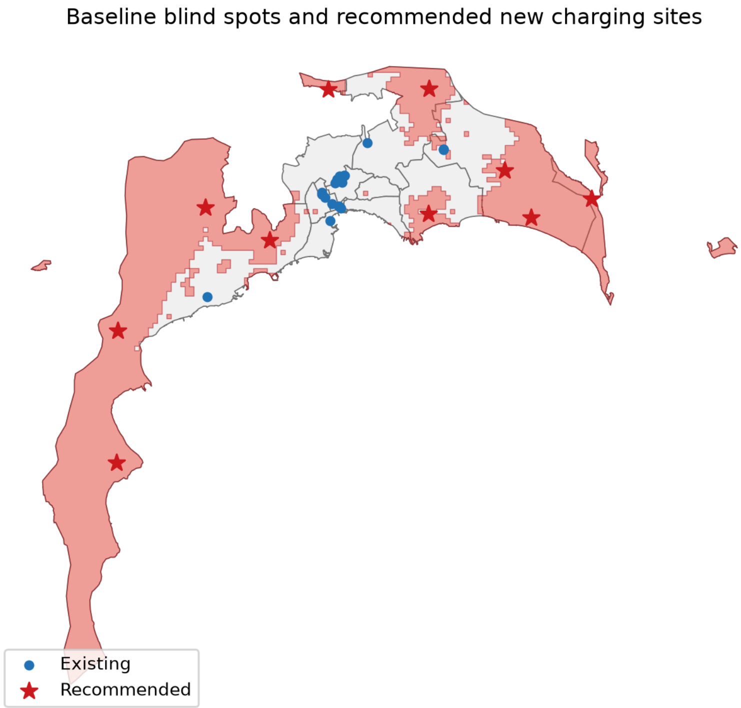

Figure 1. Baku land-clipped districts, baseline >15-minute zones, existing chargers, and recommended sites.

2. Analytical method

Accessibility surface

Baku land is partitioned into 750 m equal-area squares and clipped to district land. Each cell's representative point is snapped to the nearest drivable graph node; points farther than the configured snap tolerance are unreachable. An access penalty converts the snap distance at 30 km/h. Because a driver travels from an origin to a charger, shortest paths are computed from all charger nodes on the reversed directed graph. This respects one-way streets.

For cell i and station set S, nearest time is $m_i=\min(t_{i\square})$. Land coverage at threshold h is $100\times\Sigma\ a_i\cdot I(m_i\le h)/\Sigma a_i$. Population coverage replaces cell land a_i with WorldPop population p_i. Thresholds are 300, 600, and 900 seconds. Infinite times remain unreachable; only mean/quantile fields explicitly named capped30 cap values at 30 minutes.

Candidate sites and optimization

Candidate hosts come from OSM parking, fuel, supermarket/mall, transport-station, university, town-hall, community-centre, and marketplace features. Explicit private/no sites and POIs too near an existing charger are removed. Weighted K-means summarizes the >15-minute cells; the nearest feasible POIs to cluster centres form a bounded candidate pool. Candidate coverage is calculated with reverse cutoff Dijkstra, never straight-line buffers.

A binary maximum-coverage location model selects up to p sites. The objective is lexicographic: maximize newly covered WorldPop population at 15 minutes, then 10, then 5. Pairwise constraints enforce the minimum site spacing. SciPy's MILP/HiGHS solver is used; a documented greedy fallback is available. Scenarios p=1…10 are solved independently. The final p-site set is ordered by marginal 15/10/5-minute population gain to give an implementable deployment sequence.

Quality controls

Automated checks require $C5\le C10\le C15$; after-times never exceed baseline times; grid geometries are valid; recommended coordinates are unique; and minimum spacing is met. All API responses, inputs, parameters, package versions, selected IDs, and solver status are retained in the reproducibility package.

Model parameter	Value
Grid	750 m, clipped to land
Time thresholds	5, 10, 15 minutes
Population objective	WorldPop 2026 constrained 100 m
Candidate clusters	50
Minimum selected spacing	1500 m
Recommendation budget	10
Random seed	42

3. Baseline accessibility

At baseline, 3.9%, 16.3%, and 33.3% of analyzed land is within 5, 10, and 15 minutes of a charger. The corresponding WorldPop-weighted shares are 29.8%, 58.8%, and 84.1%. The difference between land and population results shows why a single coverage percentage can mislead: Baku includes large industrial, coastal, and low-density areas.

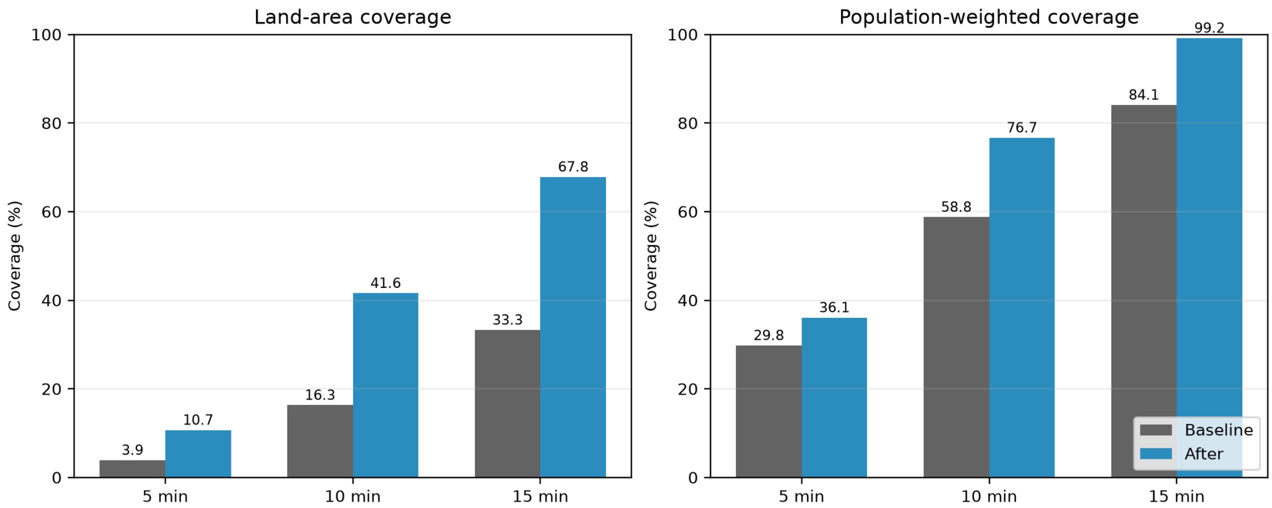

Figure 2. Land-area and population-weighted coverage at the three policy thresholds.

Lowest-accessibility districts

District	Existing	Baseline pop. ≤15	After pop. ≤15	Uplift
Pirallahı	0	0.0%	88.0%	+88.0 pp
Qaradağ	1	45.4%	96.3%	+50.9 pp
Sabunçu	1	62.3%	99.9%	+37.5 pp
Xəzər	2	65.7%	99.4%	+33.7 pp
Suraxanı	0	83.5%	99.0%	+15.5 pp
Binəqədi	5	89.4%	100.0%	+10.6 pp
Yasamal	0	97.4%	97.4%	+0.0 pp

Districts are ranked by ascending baseline population coverage within 15 minutes, with land coverage used as a secondary indicator. Charger counts are site records, not connector or plug counts; a district with one high-capacity hub and a district with one single-port charger both show one site.

4. Spatial equity and blind spots

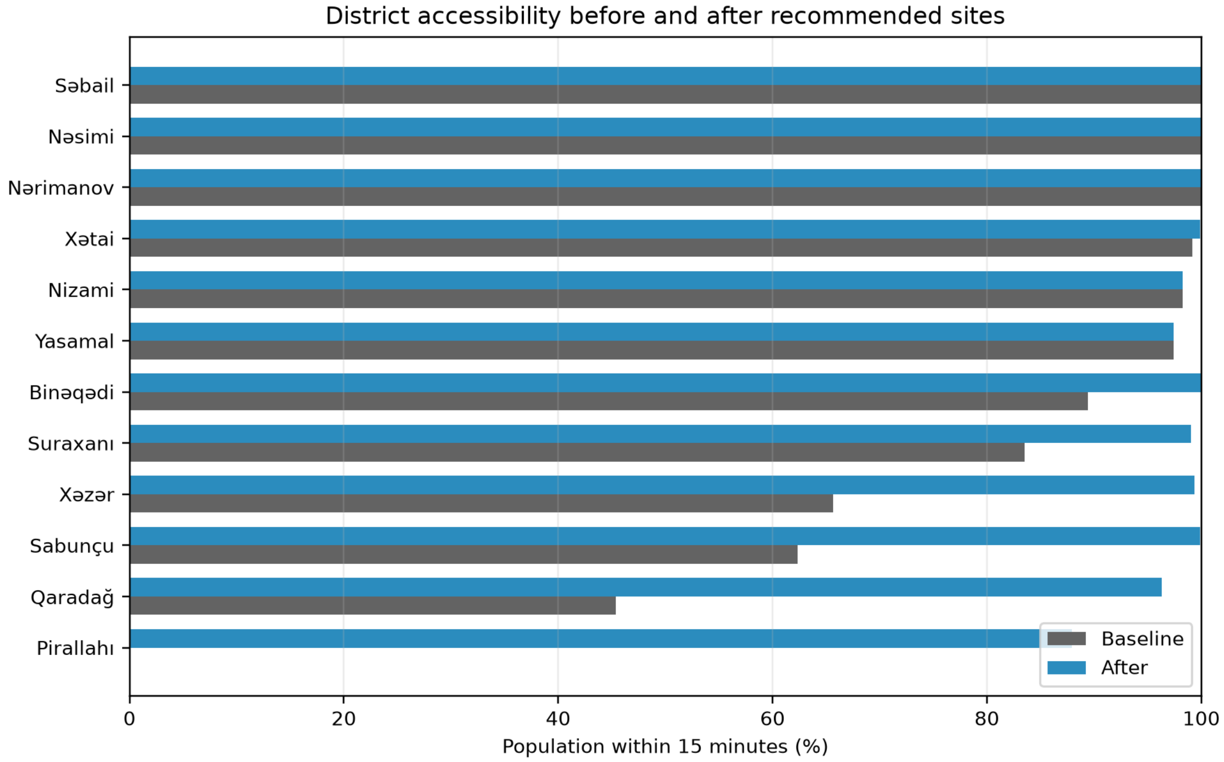

Figure 3. Population-weighted 15-minute accessibility by district before and after recommendations.

A blind spot is a contiguous set of grid cells with a modeled nearest-station time above 15 minutes or no directed route. Zones are ranked by land area multiplied by mean excess travel time (capped at 30 minutes). This ranking identifies large, severe territorial gaps; the optimization itself uses population weights.

Zone	Primary district	Area km²	Mean excess min	Severity score
BLIND-001	Qaradağ	892.4	12.0	10680.3
BLIND-002	Xəzər	249.0	10.2	2538.3
BLIND-003	Sabunçu	93.9	2.6	245.5
BLIND-004	Pirallahı	10.0	15.0	149.6
BLIND-005	Pirallahı	8.0	15.0	120.7
BLIND-006	Suraxanı	39.1	2.6	103.1
BLIND-007	Binəqədi	11.3	4.6	51.5
BLIND-008	Qaradağ	2.5	15.0	37.8

5. Recommended new charging locations

The final 10-site solution maximizes population-weighted coverage within the screened OSM candidate pool and spacing constraints. The ranking below is the recommended deployment order inside that final optimal set, not proof that every site is buildable. Marginal population is the additional WorldPop estimate brought within 15 minutes at the step when the site is added.

Rank	Site	Type	District	Lat	Lon	New pop. ≤15
1	Parkinq	parking	Sabunçu	40.55493	50.02283	153,314
2	Parking	parking	Xəzər	40.43506	50.17040	52,887
3	Azpetrol	fuel_station	Qaradağ	39.99447	49.43096	38,908
4	Parking	parking	Suraxanı	40.36946	50.02442	37,958
5	Goradil Stansiyası	transport_station	Binəqədi	40.55266	49.82808	35,883
6	Gürgan İcra Nümayəndəliyi	townhall	Pirallahı	40.39478	50.33888	26,629
7	Parking	parking	Qaradağ	40.32774	49.71809	18,072
8	Parking	parking	Qaradağ	40.19014	49.42864	13,574
9	Parking	parking	Qaradağ	40.37459	49.59310	12,240
10	Oba Market	supermarket	Xəzər	40.36533	50.22304	10,333

Implementation screening sequence

1) Confirm that each mapped POI is genuinely public or contractable and has safe 24/7 vehicle circulation. 2) Request utility hosting-capacity and connection-cost screening. 3) Verify land ownership, parking-bay control, disability access, drainage, lighting, fire safety, and permits. 4) Validate connector mix and power against the expected trip market. 5) Re-run network access with field-confirmed sites and budget/cost constraints.

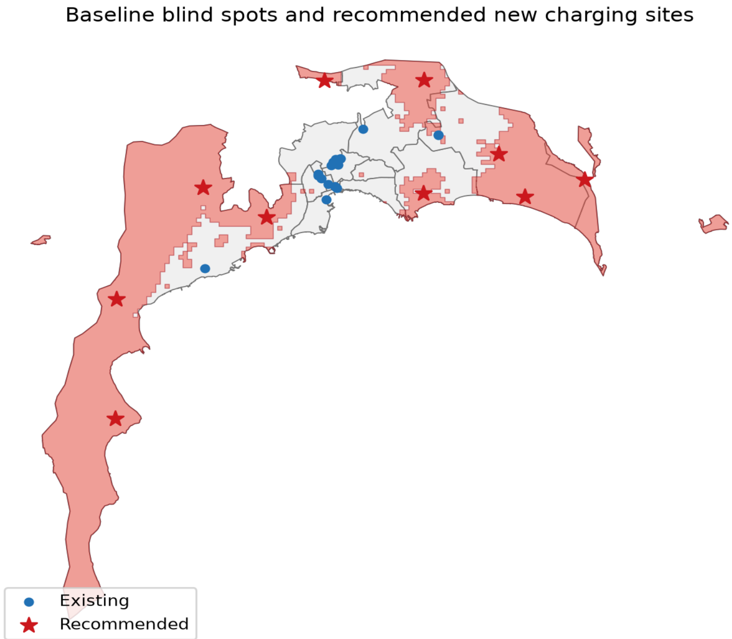

Figure 4. Geographic distribution of the final recommendation shortlist.

6. Expected improvement

With all 10 proposed sites, modeled 15-minute population accessibility rises from 84.1% to 99.2% (+15.1 percentage points). Land-area coverage rises from 33.3% to 67.8% (+34.5 points). All figures are recomputed from the minimum network time after adding sites, so overlapping service areas are not double-counted.

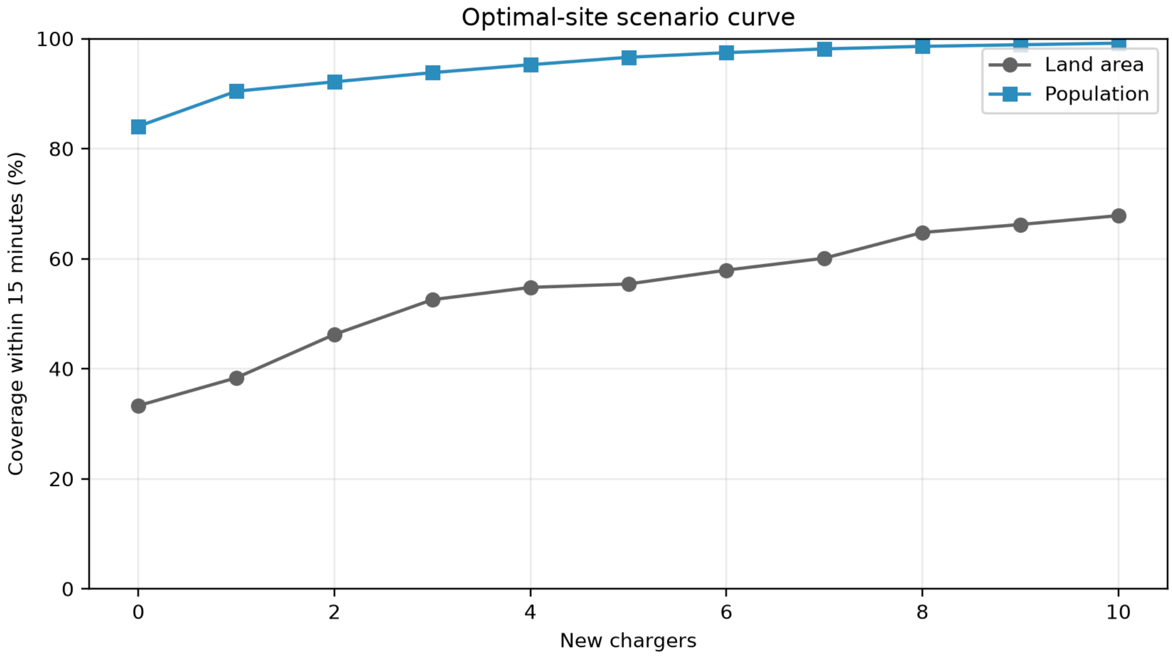

Figure 5. Independent optimal scenarios for successive site budgets. Sets need not be nested.

Measure	Baseline	After	Change
Land ≤5 min	3.9%	10.7%	+6.8 pp
Land ≤10 min	16.3%	41.6%	+25.3 pp
Population ≤5 min	29.8%	36.1%	+6.3 pp
Population ≤10 min	58.8%	76.7%	+17.9 pp
Population mean time (capped 30)	9.5 min	7.1 min	-2.4 min

7. Limitations and conclusions

Key limitations

• OSM may omit stations, retain closed sites, misclassify device chargers, or lack access, connector, power, capacity, hours, fee, and operator tags. The inclusive baseline is a reproducible scenario, not a verified operator inventory.
• OSM road speeds are static and largely free-flow; Baku congestion, incidents, turn delays, queues, and parking search are not observed.
• WorldPop R2025A is a modeled alpha population surface. It does not represent EV ownership, jobs, commuting, visitors, freight, or destination charging demand.
• A 750 m grid, road snapping, land-mask resolution, and district boundaries introduce scale and positional error. Border areas may use chargers outside the study boundary.
• Maximum coverage does not test grid capacity, capital/operating cost, land tenure, permitting, reliability, utilization, power, or connector compatibility. Recommended POIs require technical and commercial feasibility studies.

Conclusion

The analysis establishes a transparent baseline, identifies district and contiguous network blind spots, and converts those gaps into a field-screening shortlist. Its main value is comparative: it shows where a fixed number of sites provides the largest modeled accessibility gain under one consistent road, population, and boundary snapshot. Before procurement, the inclusive station inventory should be operator-verified, live or historical congestion should be tested, and the optimization should be repeated with site costs, electrical hosting capacity, forecast EV demand, and an explicit district equity constraint.

Automated verification

5/10/15-minute coverage is monotonic in every scenario; recommended sites never increase a cell's nearest-station time; recommended-site geometries are unique; recommended sites satisfy the minimum spacing rule; analysis-grid geometries are non-empty and valid.

References and data links

Azerbaijan IDDA Open Data Portal, Regions of Azerbaijan (official district polygons): https://opendata.az/en/@azerbaycan-respublikasinin-ekologiya-ve-tebii-servetler-nazirliyi/azerbaycanin-rayonlari
OpenStreetMap contributors, ODbL 1.0: https://www.openstreetmap.org/copyright
OpenStreetMap charging-station tagging: https://wiki.openstreetmap.org/wiki/Tag:amenity=charging_station
WorldPop, Azerbaijan 2026 constrained 100 m population, DOI 10.5258/SOTON/WP00839: https://hub.worldpop.org/geodata/summary?id=72400
Natural Earth 1:10m land: https://www.naturalearthdata.com/
OSMnx: https://osmnx.readthedocs.io/ · SciPy MILP: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.milp.html

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